Exercise Sheet

Lecture 13

Metric, Normed, and Hilbert Spaces

Inner products are conjugate-linear in the first slot (physics convention): $\langle x, y\rangle = \sum_n \overline{x_n} y_n$ on ℓ^2 and $\langle f, g\rangle = \int \overline{f} g$ on L^2. A Hilbert basis means a complete orthonormal family (countable only when separability is assumed), and $\mathcal{H}^*$ means the continuous dual; complex Riesz is conjugate-linear. A $\star$ marks a harder problem.

Core tutorial route

Work **13.2, 13.8, 13.11, 13.28, 13.27, 13.29, 13.21, and 13.32** in that order (about 85–100 minutes before discussion). This route covers metric versus norm and a parallelogram decision; a concrete incomplete space and completion; measure-aware ℓ^2/L^2 reasoning; basis completeness versus ambient completeness; the Bessel–expansion proof spine; closed-subspace best approximation; one justified Fourier bridge; and bounded complex Riesz.

Additional practice: 13.1, 13.3–13.7, 13.10, 13.12–13.19, 13.22–13.23, 13.26, 13.30–13.31, 13.33, and 13.35. **Bridge:** 13.20 and 13.24–13.25 (Fourier modes and approximation). **Extension:** 13.9 (absolute-series completeness) and 13.34 (evaluation spikes). Bridge and extension tasks do not count toward core progress.

Metrics, norms, completeness

Exercise 13.1 (proof) Show that $d(x, y) = \|x - y\|$ satisfies the triangle inequality whenever $\|\cdot\|$ is a norm.

Exercise 13.2 (proof) On $C[0, 1]$ let $\|f\|_\infty = \sup_x |f(x)|$ and $d(f, g) = \|f - g\|_\infty$. Explain why d is induced by a norm, then decide whether that norm comes from an inner product by testing $f = 1$ and $g(x) = x$.

Exercise 13.3 (computational) For $x = (2, -1, 2) \in \mathbb{R}^3$ compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Exercise 13.4 (computational) For $x = (1, -2, 2, -4) \in \mathbb{R}^4$ compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Exercise 13.5 (computational) For $x = (3i, -4) \in \mathbb{C}^2$ compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Exercise 13.6 (computational) Does the taxicab norm $\|\cdot\|_1$ on $\mathbb{R}^2$ come from an inner product? Test the parallelogram law with $u = (1, 0)$, $v = (0, 1)$.

Exercise 13.7 (computational) Verify the parallelogram law for the Euclidean norm on $\mathbb{R}^2$ with $u = (3, 0)$, $v = (0, 4)$, consistent with $\|\cdot\|_2$ coming from an inner product.

Exercise 13.8 (proof) Let q_n be the decimal truncation of π after n digits, regarded as an element of $\mathbb{Q}$ with the usual metric. Show that (q_n) is Cauchy in $\mathbb{Q}$ but does not converge there. Name its completion and its limit in that completion.

Exercise 13.9 ($\star$) **Optional extension.** Prove that a normed space is complete if and only if every absolutely convergent series converges.

Exercise 13.10 (proof) Prove the reverse triangle inequality $\big|\, \|u\| - \|v\| \,\big| \le \|u - v\|$ in any normed space.

ℓ^2 and L^2

Exercise 13.11 (proof) Answer both parts.

1. Is $x_n = 1/n$ in ℓ^2? Is $x_n = 1/\sqrt{n}$?
2. In $L^2[0,1]$, let $f = 0$ and let g equal 1 at $x = 0$ and 0 elsewhere. Explain why f and g are the same L^2 vector and why this prevents point evaluation at 0 from being a functional on L^2.

Exercise 13.12 (computational) Compute $\langle f, g\rangle$ for $f(x) = 1$, $g(x) = x^2$ on $L^2[0,1]$, and $\|g\|$.

Exercise 13.13 (computational) Normalize $\psi(x) = e^{-x}$ on $[0,\infty)$ in L^2.

Exercise 13.14 (computational) For $x = (1, \frac{1}{2}, \frac{1}{4}, \dots) \in \ell^2$ (that is $x_n = 2^{-n}$, $n \geq 0$) compute $\|x\|$.

Exercise 13.15 (computational) For $x_n = 2^{-n}$ and $y_n = 4^{-n}$ $(n \geq 0)$ in ℓ^2, compute $\langle x, y\rangle$.

Exercise 13.16 (computational) For which real r is $(1, r, r^2, r^3, \dots) \in \ell^2$, and what is its norm then?

Exercise 13.17 (computational) Compute $\langle f, g\rangle$ for $f(x) = x$, $g(x) = x^2$ on $L^2[0,1]$.

Exercise 13.18 (computational) Normalize $\psi(x) = x^2$ on $[0,1]$ in L^2.

Exercise 13.19 (proof) Prove that if $x = (x_n) \in \ell^2$ then $x_n \to 0$. Show by example that the converse fails.

Orthonormal bases and Fourier series

Exercise 13.20 (computational) The functions $\frac{1}{\sqrt{2\pi}}e^{inx}$ are orthonormal on $L^2[-\pi,\pi]$. Verify orthonormality for $n = 0$ and $n = 1$.

Exercise 13.21 (proof) **Bridge task within the core route.** The L^2 Fourier calculation for x^2 on $[-\pi,\pi]$ gives

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n\geq 1}\frac{(-1)^n}{n^2}\cos(nx) \quad \text{in } L^2.$$

Explain why the series converges uniformly, why its continuous limit must equal x^2 at every point, and only then evaluate at 0 and π to find the alternating and ordinary sums of reciprocal squares.

Exercise 13.22 (computational) On $L^2[-1,1]$ show that 1 and x are orthogonal, and compute $\|1\|$ and $\|x\|$.

Exercise 13.23 (computational) Normalize 1 and x on $L^2[-1,1]$ to obtain the orthonormal pair e_0, e_1.

Exercise 13.24 (computational) Verify orthonormality of $e_n = \frac{1}{\sqrt{2\pi}}e^{inx}$ on $L^2[-\pi,\pi]$ for the indices $n = 2, 3$: compute $\langle e_2, e_2\rangle$ and $\langle e_2, e_3\rangle$.

Exercise 13.25 (computational) Find the best approximation of $f(x) = x$ on $L^2[-\pi,\pi]$ by the orthogonal set $\{\sin x, \sin 2x\}$.

Exercise 13.26 (computational) Let $\{e_1, e_2, e_3\}$ be orthonormal in a Hilbert space and v satisfy $\|v\|^2 = 10$ with $\langle e_1, v\rangle = 1$, $\langle e_2, v\rangle = 2$, $\langle e_3, v\rangle = 2$. Compute $\|Pv\|$, where P projects onto $\operatorname{span}\{e_1, e_2, e_3\}$, and the squared distance $\|v - Pv\|^2$; check Bessel's inequality.

Exercise 13.27 (proof) Let (e_n) be an orthonormal sequence in a Hilbert space and $s_N = \sum_{n\le N}\langle e_n, v\rangle e_n$. Prove the expansion spine: finite Pythagoras gives Bessel; coefficient tails make (s_N) Cauchy; ambient Hilbert completeness gives a limit s; and $v - s$ is orthogonal to the closed span. Conclude that Parseval equality is equivalent to completeness of the family.

Exercise 13.28 (proof) Separate these two meanings of "complete." Let c_{00} be the finite-support sequences with the ℓ^2 norm, and let (e_n) be the standard coordinate family.

1. Show (e_n) has dense span in both c_{00} and ℓ^2.
2. Explain why the family is complete in that basis sense in both spaces, while only the ambient space ℓ^2 is complete as a metric space.

Projection and Riesz

Exercise 13.29 (computational) Find the orthogonal projection of $f(x) = x$ onto the constants in $L^2[0,1]$, and the residual.

Exercise 13.30 (proof) Let $m(0) = 1$ and $m(x) = 0$ for $x \ne 0$ on $[0,1]$. Find the pointwise supremum, essential supremum, and norm of the multiplication operator M_m on $L^2[0,1]$. Explain what this tests.

Exercise 13.31 (computational) Find the orthogonal projection of $f(x) = x^3$ onto $\operatorname{span}\{1, x\}$ in $L^2[-1,1]$.

Exercise 13.32 (proof) On complex ℓ^2 define $\varphi(x) = \sum_{n\ge1} i x_n/3^n$. Prove that it is bounded, find the Riesz vector u with $\varphi = \langle u, \cdot\rangle$, and verify directly that the Riesz map $R(u) = \langle u, \cdot\rangle$ satisfies $R(\alpha u) = \overline{\alpha}R(u)$.

Exercise 13.33 (computational) By Riesz, find $u \in L^2[0,1]$ with $\varphi(f) = \langle u, f\rangle$ for the functional $\varphi(f) = \int_0^1 (2+3x)\, f(x)\, dx$.

Exercise 13.34 (⋆) **Optional extension.** On $C[0,1]$ equipped with the L^2 norm, define $f_n(x) = \max(1 - nx, 0)$. Show that point evaluation $E(f) = f(0)$ is linear and well-defined there but unbounded. Explain why the same formula is not even well-defined on the quotient space $L^2[0,1]$.

Exercise 13.35 (proof) Show that the vector u representing a bounded functional via Riesz is unique: if $\langle u, \cdot\rangle = \langle u', \cdot\rangle$ as functionals, then $u = u'$.